

Dayi Fan

PERSONAL INFO

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RESEARCH INTEREST

My research focuses on parallel computing, algorithm engineering, and scalable data systems. Specifically, I design novel parallel algorithms and develop hardware-aware implementations on multicore CPUs and GPUs to accelerate irregular workloads across graph analytics, database systems, and other data-intensive applications. My work has produced state-of-the-art parallel solutions for several fundamental problems: tree edit distance, graph maximum matching, and bipartite weighted matching.

My future research agenda also extends to ML systems and AI agents. I have hands-on experience with Transformers and Hugging Face libraries, from model training to inference deployment, as well as with OpenEvolve for AI-guided code evolution. In addition, I have extensive experience designing and building automated research workflows that integrate and coordinate multiple AI models, including Gemini, Claude, and Codex. I have applied these workflows in my recently submitted CGO papers. My vision is to make computing systems more automatic, adaptive, and resource-efficient.

EDUCATION

2022.01-2026.12 (expected)	Ph.D. Candidate in Computer Science and Engineering, The Ohio State University Advisor: Prof. Xiaodong Zhang Theme: Parallel Algorithm Design and Hardware Acceleration for Data-Centric Applications
2021.01-2021.05	Undergraduate Exchange Student in Computer Science, National University of Singapore Project: LSTM Implementation with Spiking Neural Networks

2017.09-2021.07 B.Eng. in Microelectronics Science and Engineering,
Southern University of Science and Technology

PAPERS IN SUBMISSION

2026 Veray: A Language for Repurposed Ray Tracing
Submitted to CGO 2027

- A restricted-Python DSL that *automatically* generates GPU code and execution paths for repurposed ray-tracing applications
- Designed and optimized multi-stage automated research workflows that orchestrate multiple AI agents, including Gemini, Claude, and Codex.

PUBLICATIONS

2026 X-Wim: Massive Parallelization of Weighted Matching in Bipartite Graphs
Dayi Fan, Simon Zhang, Rubao Lee, Hanqi Guo, Xiaodong Zhang
Proceedings of the VLDB Endowment (VLDB 2026) [[PDF](#)][[Code](#)]

- The first scalable parallel framework for bipartite weighted matching, a fundamental problem in user paring, object tracking, and entity resolution
- Introduces a new phase-decoupled Hungarian algorithm that breaks the tight interleaving between path search and label updates, enabling massive search concurrency while substantially reducing synchronization
- Outperforms state-of-the-art baselines by up to 10x on one core and scales to 96 cores with an average 1.7x speedup per thread doubling

2025 X-Blossom: Massive Parallelization of Graph Maximum Matching
Dayi Fan, Rubao Lee, Xiaodong Zhang
Proceedings of the VLDB Endowment (VLDB 2025) [[PDF](#)][[Code](#)]

- The first scalable parallel framework for maximum matching, a critical primitive in graph databases and quantum error correction.
- Proposes a new recursion-free Blossom algorithm that eliminates dynamic graph contraction, thereby removing strong recursive dependencies and exposing massive parallelism
- Surpasses prior state-of-the-art matching solutions significantly, achieving speedups of up to 992x

- 2024 **X-TED: Massive Parallelization of Tree Edit Distance**
Dayi Fan, Rubao Lee, Xiaodong Zhang
Proceedings of the VLDB Endowment (VLDB 2024) [[PDF](#)][[Code](#)]
- The first scalable parallel framework for tree edit distance, with broad applications including code analysis and LLM output evaluation
 - Develops a new algorithm that infers intricate data dependencies among dynamic-programming tables from input tree structures with negligible overhead and batches independent tables for concurrent processing
 - Achieves speedups of up to 31× over prior parallel solutions
- 2024 **RR-Compound: RDMA-Fused gRPC for Low Latency, High Throughput, and Easy Interface**
Liang Geng, Hao Wang, Jingsong Meng, **Dayi Fan**, Sami Ben-Romdhane, Hari Kadayam Pichumani, Vinay Phegade, Xiaodong Zhang
IEEE Transactions on Parallel and Distributed Systems (TPDS 2024) [[PDF](#)][[Code](#)]
- Integrates RDMA into gRPC with hybrid connection management and preallocated buffers to reduce latency and improve throughput
 - Outperformed standard gRPC over IPoIB, delivering up to 2.35× speedup for key-value storage and 1.3× for distributed TensorFlow workloads

INVITED TALKS

Aug 31, 2026 **Massive Parallelization of Irregular Computations:
Parallel Algorithm Design and Hardware Acceleration** [[Talk](#)]
MIT Computer Science and Artificial Intelligence Laboratory (CSAIL)
Cambridge, MA · Room 32-G575 · 2:00–3:00 PM
Hosted by Prof. Julian Shun

ACADEMIC SERVICE

- 2027 Program Committee Member, ACM SIGMOD 2027
- 2027 Shadow Program Committee Member, VLDB 2027

2026 Session Chair, VLDB 2026

2026 Reviewer, The Journal of Supercomputing

EXPERIENCE

2022.01-CURRENT Graduate Research Associate
The Ohio State University

2022.01-2024.05 Graduate Teaching Associate
The Ohio State University
CSE 3421 Computer Architecture
OSU C4 Oral English Proficiency Certification

SKILLS

Programming Languages: C++, CUDA, Python, Java

Databases: SQL, PostgreSQL, MySQL

Machine Learning: PyTorch, Transformers, Hugging Face, OpenEvolve

Systems/HPC: Multithreading, GPU Programming, System Profiling, gRPC, RDMA

Multi-Agent Workflow Orchestration: Gemini, Claude, Codex, DeepSeek

ACADEMIC PERFORMANCE

Ph.D. GPA: 3.96/4.00

2021.06 Excellent Graduation Award
Southern University of Science and Technology

2017-2020 First Class Student Scholarship (three consecutive years)
Southern University of Science and Technology